Cobalt CX-22

Centrifugal Pump - Synthetic Maintenance Guide

ASSET REFERENCE	SYN-CX-022
DOCUMENT STATUS	Revision 2.0 / Synthetic edition
DATA CLASS	Entirely fictional / non-production

1. Purpose and equipment summary

The fictional Cobalt CX-22 transfers clean synthetic process water. The test assembly includes an inlet strainer, mechanical seal flush, flexible coupling, and two radial vibration measurement points.

This manual exists only to exercise document upload, PDF extraction, page provenance, retrieval, and citation behavior in QIP. It does not describe real equipment and must not be used for physical maintenance.

Nominal operating envelope

Parameter	Synthetic value
Nominal flow	22 m3/h
Rated speed	2,900 rpm
Minimum inlet pressure	0.40 bar gauge
Seal-flush target	1.6-2.0 L/min
Vibration alert	4.5 mm/s RMS
Immediate shutdown	7.1 mm/s RMS

2. Alarm and troubleshooting reference

Preserve observations before resetting an alarm. The table provides decision support for synthetic investigations; it does not establish root cause by itself.

V21	**Broadband vibration above 4.5 mm/s RMS**

Recommended inspection: Check inlet pressure and strainer condition before alignment. Cavitation is likely when inlet pressure is below 0.40 bar and noise resembles gravel.

S04	**Seal-flush flow below 1.6 L/min**

Recommended inspection: Inspect the flush orifice and isolation valve. Restore 1.6-2.0 L/min before prolonged operation.

B09	**Drive-end bearing temperature above 82 C**

Recommended inspection: Stop at 90 C. Inspect lubrication quantity and coupling alignment after lockout.

INVESTIGATION EXERCISE	Scenario CX-B: Operators report gravel-like noise and 5.2 mm/s vibration. Inlet pressure is 0.28 bar, while seal-flush flow remains 1.8 L/min. Evidence supports inlet restriction and cavitation as the leading hypothesis; alignment should not be adjusted first.

3. Inspection and maintenance

Use the following synthetic intervals to test evidence capture, question answering, and citation to a specific page.

- Daily: record inlet pressure, discharge pressure, seal-flush flow, and drive-end vibration.
- Every 500 synthetic hours: inspect coupling insert and clean the inlet strainer when pressure loss exceeds 0.18 bar.
- Every 2,000 synthetic hours: inspect bearing lubricant and verify sensor readings against a portable reference.
- After a 7.1 mm/s shutdown: preserve readings and inspect for cavitation damage before restart.

Safety boundary

TEST DATA ONLY	Lockout, isolation, and competent-person requirements in this fictional document are illustrative. Never apply these values or procedures to real machinery.

Suggested QIP test questions

- What observation should be preserved before changing a machine setting?
- Which threshold in the source supports the recommended first inspection?
- Is there enough evidence to state a confirmed root cause? Explain the uncertainty.